

Incident handler's journal

Date: October 17, 2026	**Entry:** **Entry 1 — Healthcare Clinic Ransomware Security Incident**
Description	Incident log documenting a ransomware attack at a primary-care clinic caused by targeted email phishing, resulting in system encryption and operational shutdown.
Tool(s) used	Email security filters, Endpoint Detection and Response (EDR) software, SIEM/log analysis tools, Data backup and restoration software.
The 5 W's	T**he 5 W's** • **Who caused the incident?** An organized threat actor group of unethical hackers known for targeting healthcare and transportation sectors, using unsuspecting clinic employees who opened malicious phishing attachments as the initial entry point. • **What happened?** A ransomware attack took place. Attackers delivered malware via targeted phishing emails with malicious attachments. Once downloaded, the malware allowed attackers to access the network and deploy ransomware, encrypting critical medical records and displaying a ransom note demanding payment for the decryption key. • **When did the incident occur?** On a Tuesday morning at approximately 9:00 a.m. • **Where did the incident happen?** At a small U.S. healthcare clinic specializing in primary-care services, impacting the clinic's internal network, employee workstations, and database servers housing patient records. • **Why did the incident happen?** The threat actors conducted the attack for financial gain via ransom extortion. The breach succeeded because

	targeted phishing emails bypassed initial email filters and enticed employees to download malicious attachments, exploiting a lack of robust endpoint security controls.
Additional notes	• **Impact:** Complete disruption of business operations and loss of access to critical patient data, forcing a total computer system shutdown and external incident escalation for technical and legal support. • **Immediate Action Items:** Isolate compromised endpoints from the network, preserve malware samples and email headers for forensic analysis, assess backup integrity to restore files without paying the ransom, and notify relevant regulatory bodies (e.g., HHS/HIPAA breach notification authorities). • **Long-Term Preventative Controls:** Implement advanced email filtering/sandboxing, conduct mandatory employee phishing awareness training, enforce strict endpoint protection (EDR), and maintain offline, air-gapped data backups.

Date: October 18, 2026	**Entry:** Entry 2 — Suspicious File Download Alert
Description	Investigated a SOC alert regarding an employee downloading and executing a password-protected spreadsheet attachment, which resulted in malicious payloads being dropped.
Tool(s) used	VirusTotal, Intrusion Detection System (IDS)

The 5 W's	• **Who caused the incident**? An external threat actor using spearphishing. • **What happened?** An employee received a password-protected spreadsheet via email, downloaded it, and entered the provided password. This executed a malicious payload that dropped unauthorized executable files. • **When did the incident occur?** At 1:11 p.m. (email received), 1:13 p.m. (file executed), and 1:20 p.m. (IDS alert generated). • **Where did the incident happen?** On an employee's computer within the financial services company network. • **Why did the incident happen?** The attacker used defense evasion (a password-protected file) to bypass email security scanners and relied on user execution to deliver the payload.
Additional notes	VirusTotal analysis of the SHA256 hash (54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b) confirmed it as a malicious Trojan with a high vendor detection ratio. Key IoCs identified include the MD5 hash (287d612e29b71c90aa54947313810a25), contacted IP (207.148.109.242), the associated malicious domain, and dropped host artifacts (e.g., K80K8X6S.htm).

Date:	**Entry:**
October 19, 2026	Entry 3 — Investigating Network Traffic with Suricata

Description	Investigated a security alert triggered by Suricata, an open-source intrusion detection system, concerning suspicious HTTP traffic requesting index.php. This falls under the Detection and Analysis phase of the NIST Incident Response Lifecycle, focusing on verifying malicious network activity.
Tool(s) used	• Suricata (Intrusion Detection System) • Linux command line (cat, ls, grep) for log analysis
The 5 W's	• **Who caused the incident?** An external threat actor utilizing the IP address 172.21.224.2. • **What happened?** Suricata rules flagged an incoming HTTP GET request for a potentially vulnerable file (index.php) as malicious, generating an alert in the fast.log and detailed payload information in eve.json. • **When did the incident occur?** The event was logged at 04:32:00 UTC based on the Suricata timestamp. • **Where did the incident happen?** The attack originated from an external host (172.21.224.2) and targeted the internal web server located at 172.21.224.1 on port 80. • **Why did the incident happen?** The attacker was attempting to probe or exploit potential vulnerabilities within the web server application, likely scanning for unsecured administrative interfaces or executable PHP files.
Additional notes	Reviewing eve.json confirmed the alert was valid (True Positive) and provided additional context about the HTTP request headers. The Suricata rule successfully identified the malicious signature.

Date: October 20, 2026	**Entry:** Entry 4 — Splunk Query for Failed SSH Logins
Description	Performed a query using Splunk to investigate multiple failed SSH login attempts originating from an external IP address targeting an internal mail server. This occurred during the Detection and Analysis phase to determine the scope of a potential brute-force attack.
Tool(s) used	• Splunk (Security Information and Event Management) • Splunk Processing Language (SPL)
The 5 W's	• **Who caused the incident?** An unknown attacker launching an automated attack from the external IP address 192.168.1.50. • **What happened?** A high volume of failed SSH login attempts was recorded, indicating a likely dictionary or brute-force attack attempting to guess administrative credentials. • **When did the incident occur?** The failed login spikes occurred over a short time frame (e.g., between 02:00 and 02:15 system time) as visualized on the Splunk timeline. • **Where did the incident happen?** The login attempts were directed at the internal SSH daemon (port 22) on the mail server (10.0.0.15). • **Why did the incident happen?** The attacker was attempting to gain unauthorized remote access to the internal network by exploiting weak or default passwords on exposed services.
Additional notes	The Splunk query (index=main sourcetype=linux_secure "Failed password for") effectively filtered the logs to show only the relevant failed attempts. The attacker failed to gain access; however, IP blocking should be considered for the attacking address to mitigate further attempts.

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Tool(s) used	List any cybersecurity tools that were used.
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Need another journal entry template?

If you want to add more journal entries, please copy one of the tables above and paste it into the template to use for future entries.

Were there any specific activities that were challenging for you? Why or why not?
Navigating log structures in tools like Splunk and Chronicle was initially challenging because the syntax requires precision. However, breaking down the search queries into smaller, verifiable steps made it easier to extract meaningful security events from large datasets.

Has your understanding of incident detection and response changed since taking this course?
Yes, my understanding has expanded significantly. I now recognize that incident response is a structured lifecycle (like the NIST framework) rather than just reacting to alerts, and proper documentation is just as important as technical remediation.

Was there a specific tool or concept that you enjoyed the most? Why?
I enjoyed working with Suricata and analyzing packet captures. Seeing exactly how a malicious payload triggers a rule signature made abstract concepts like network intrusion detection feel practical and tangible.